

What year did *you* come from?: Predicting origin date of book & movie text

Motivation

Over the years, language has changed dramatically. We do not speak like Shakespearean plays, and two decades ago, nobody knew what an oomf was. This change in language is likely reflected in books and movies of the time: the dialogue in books from two centuries ago should have differences from contemporary books. This project uses NLP to see if the origin date of book text and movie dialogue can be predicted.

Data

- Movie dialogue was sourced from the Cornell Movie Dialogs Corpus, which contains scripts from 617 different movies.
- Book text was sourced from the Harvard Library Public Domain Corpus, with 46,449 copyrighted books and 22,548 out of copyright books.
- With selections from both sources, the final training set included 70,975 examples.

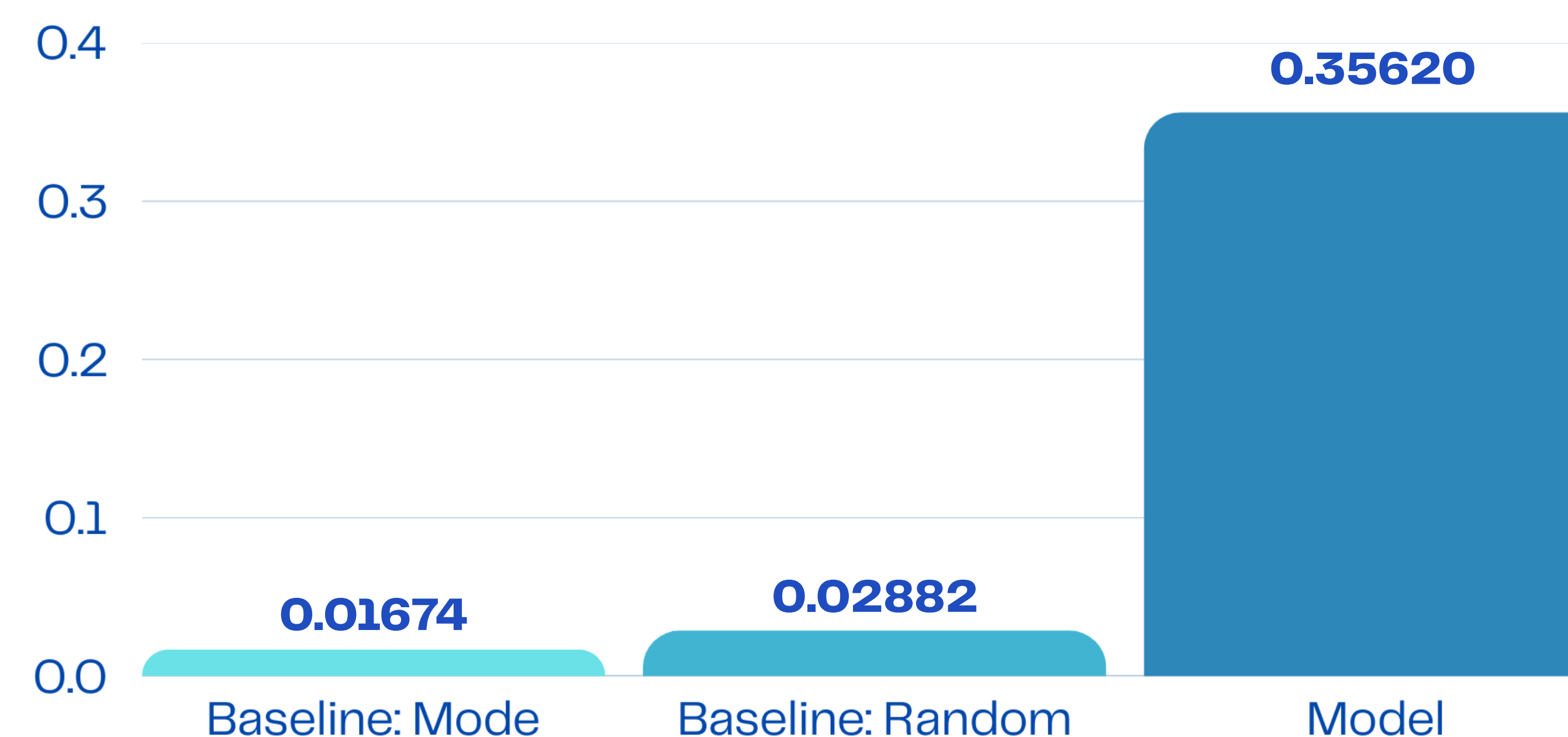
Methods

- Data was grouped into decades, with the model learning the correct decade classification.
- The model was an ordinal regression model that added an extra linear head on top of the pretrained distilbert-base-uncased model.
- The loss function is inspired by CORN and helps the model learn the order of the classes by making it a binary classification problem.
- Ran for 22,000 steps.

The model showed a significant improvement over baselines

- Performance was evaluated using the macro F1 score on a test set of 19,842 examples.
- Baselines used for evaluation included:
 - Mode: Picking the most frequent class every time
 - Random: Picking a random class every time.
- However, mean squared error was 10.5, indicating it was on average only within a century of the right decade.

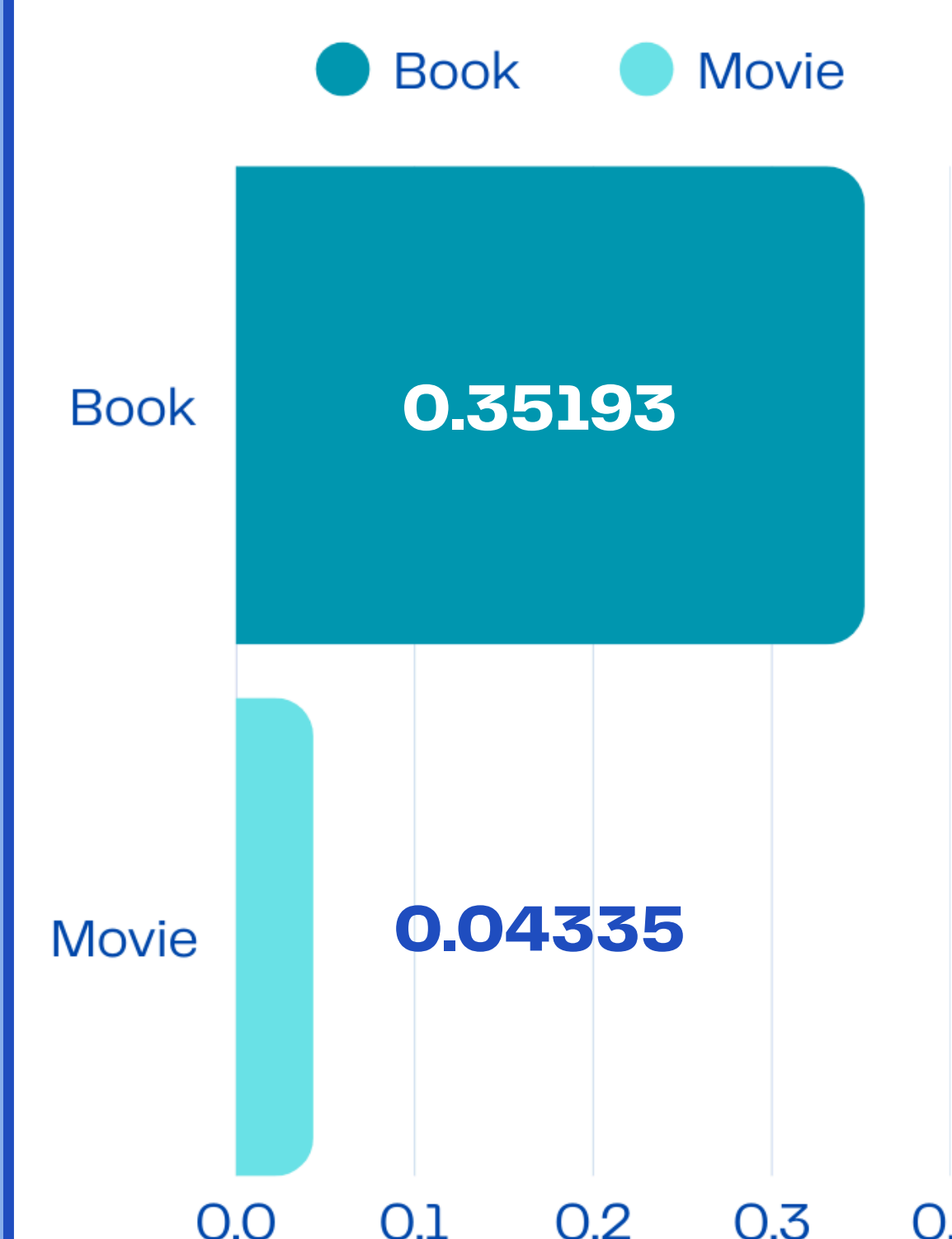
Macro F1 Score



The model was significantly better at predicting origin date of book text versus movie dialogue



Macro F1 Score
Book vs Movie Text



It is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want of a wife

Predicted:
1900
Actual:
1813

Look, I didn't want to be a half-blood. If you're reading this because you think you might be one, my advice is: close this book right now. Believe whatever lie your mom or dad told you about your birth, and try to lead a normal life.

Predicted:
1990
Actual:
2005

Conclusion

- Successfully improved upon the baseline when predicting the decade of origin of text
- But there is room for improvement:
 - Public domain corpus included many non-fiction records and publications
 - Many more book training examples than movie training examples
 - Include data from other media: comic book dialogue, music lyrics, poems
 - Run for more epochs
 - Test different hyperparameters
 - Experiment with different models
- Could also evaluate with a margin of error for the decade class